PANDUAN OPERASIONAL

Instalasi Server Baru

Dari server Ubuntu kosong sampai layanan chat & komunitas bisa diakses publik lewat domain dengan HTTPS.

Ringkasnya, tiga langkah

1. Dari laptop, kirim script ke server: `scp scripts/server-install.sh <user>@<IP-server>:~/install.sh`
2. Masuk ke server (`ssh <user>@<IP-server>`), jalankan `bash install.sh` , dan jawab pertanyaannya
3. Selesaikan pengaturan di Rendezvous Hub (bagian 5)

Sumber: docs/server-install.md · Dibuat 4 November 2026

Panduan memasang server sosial di server Ubuntu baru sampai API bisa diakses publik lewat domain dengan HTTPS. Sebagian besar langkah dikerjakan satu script, `scripts/server-install.sh`, yang dikirim ke server dengan `scp` sebelum repo di-clone di sana. Isinya juga dimuat di Lampiran.

Hasil akhirnya:

- `https://<domain API>` untuk aplikasi OLT (REST, Realtime, login, unggah media), sertifikat Let's Encrypt otomatis.
- Rendezvous Hub di `https://<domain Hub>`, hanya dari IP kantor/VPN (opsional).
- Backup berjalan ke folder di disk terpisah setelah diatur dari Hub.

Rujukan lain: operations.md (operasional), runbook-gangguan.md (gangguan), PRODUCTION.md (detail produksi), dev-server.md (spek dan mode dev).

1. Yang harus ada sebelum mulai

	Dari	Contoh
Server Ubuntu 24.04, 4 vCPU / 8 GB / 100 GB SSD (+ disk backup terpisah, mis. `/mnt/backup`)	Vendor IT	—
IP publik server (atau port forwarding 80/443 dari IP publik kantor)	Vendor IT	`203.0.113.10`
A record domain API ke IP itu	Vendor IT / pengelola DNS	`social-api.perusahaan.co.id`
A record domain Hub ke IP yang sama (opsional)	Vendor IT / pengelola DNS	`hub.perusahaan.co.id`
Firewall penyedia: 80 dan 443 dari internet, 22 hanya dari IP admin	Vendor IT	—
User SSH dengan `sudo`, login dengan SSH key	Vendor IT	—
JWKS (kunci **publik**) penanda tangan token sosial	Tim backend OLT (olt-integration.md)	`{"keys":[{"kty":"OKP",...}]}`
Email tim IT (notifikasi sertifikat)	Anda	`it@perusahaan.co.id`
IP publik kantor / subnet VPN untuk Hub	Tim IT	`198.51.100.7/32`
Akses admin repo GitHub `chat_service_supabase`, dan `rendezvous_hub` bila Hub dipasang (untuk menambah deploy key)	Anda	—

Tanpa JWKS dari OLT, mode produksi tidak bisa dipakai (kunci dummy ditolak). Selama JWKS belum ada, pakai mode dev di bagian 7.

2. Menjalankan script

1. Kirim script ke server. Cara paling aman: dari laptop yang sudah punya repo ini (PowerShell atau terminal di folder repo):

   ```
   scp scripts/server-install.sh <user>@<IP-server>:~/install.sh
   ```

   Cara lain bila repo tidak ada di laptop: setelah masuk ke server (langkah 2), jalankan `nano install.sh`, tempel isi file dari GitHub, lalu Ctrl+O, Enter, Ctrl+X. Jangan menyalin script dari PDF: baris panjang yang terlipat bisa ikut tersalin sebagai baris baru dan merusak script.

2. Masuk ke server:

   ```
   ssh <user>@<IP-server>
   ```

3. Jalankan di server:

   ```
   bash install.sh
   ```

Script menanyakan beberapa hal dan berhenti dengan pesan jelas bila ada yang belum siap. Aman dijalankan ulang: langkah yang sudah selesai dilewati, isian sebelumnya menjadi nilai bawaan.

Langkah	Yang dilakukan	Yang ditanyakan
1	Update sistem, paket dasar, update keamanan otomatis, NTP	—
2	Pasang Docker dan compose dari repo resmi Docker	—
3	Swap 4 GB bila RAM < 8 GB	—
4	`ufw` : SSH, 80, 443	—
5	Deploy key GitHub (baca saja), clone repo ke `~/chat_service_supabase`	Tambahkan deploy key yang ditampilkan di GitHub, lalu Enter
6	`make gen-secrets` , isi `docker/.env`	Domain API, email, domain Hub dan IP kantor (opsional), folder backup
7	Bila domain Hub diisi: deploy key kedua (`~/.ssh/github_deploy_hub` , alias SSH `github-hub`), clone `rendezvous_hub` ke `~/rendezvous_hub` ; saat dijalankan ulang cukup `git pull`	Tambahkan deploy key kedua di repo `rendezvous_hub` , lalu Enter

Langkah	Yang dilakukan	Yang ditanyakan
8	Simpan JWKS OLT ke `docker/secrets/olt_jwks`, tolak kunci dev atau kunci privat	Tempel JWKS, Enter, Ctrl+D
9	Cek DNS domain mengarah ke server	Konfirmasi IP
10	`make up` profil `single`, akun Super Admin pertama, verifikasi HTTPS	Email dan nama Super Admin

GitHub menolak satu deploy key dipakai di dua repo, jadi Hub memakai kunci sendiri. Image Hub dibangun dari `~/rendezvous_hub` (letaknya bersebelahan dengan `~/chat_service_supabase`). Bila `HUB_APP_DIR` diatur ke folder lain, script mengisi `HUB_DIR` di `docker/.env`.

Di akhir, script mencetak URL API, anon key, dan alamat Hub. Password sementara Super Admin hanya tampil **sekali**; catat sebelum menutup terminal.

Waktu: ±15–30 menit, sebagian besar untuk mengunduh dan build image pertama kali.

3. Isi script

Isi lengkap script dimuat di **Lampiran** di akhir dokumen ini untuk dibaca. Untuk dipakai, kirim file `scripts/server-install.sh` dari repo dengan `scp` (bagian 2), jangan disalin dari PDF.

4. Cek setelah selesai

Script sudah memeriksa tiga hal di bawah. Ulangi kapan saja:

```
cd ~/chat_service_supabase
make ps                                                    # semua healthy
curl -s -H "apikey: $(cat docker/secrets/anon_key)" https://<domain API>/bridge/v1/health
                                                           # "olt_mode":"jwt", "status":"ok"
curl -sI http://<domain API>/ | head -1                    # 301/308 ke https
curl -s -o /dev/null -w '%{http_code}\n' https://<domain API>/   # 404: hanya path API yang dibuka
```

Dari laptop di luar kantor: `nc -zv <IP-server> 5432` harus gagal, dan `https://<domain Hub>` harus 403.

5. Setelah server jalan

1. **Login Hub** (`https://<domain Hub>`, dari kantor/VPN) dengan email Super Admin dan password sementara, lalu ganti password.

2. **Pengaturan → Backup**: pilih tujuan (Folder lokal di server, atau S3), buat password enkripsi, simpan. Tekan **Uji restore sekarang** setelah backup penuh pertama selesai.
3. **Pengaturan → Alert**: isi Telegram/email, lalu **Kirim uji**. Tanpa kanal ini, alert gangguan tidak terkirim ke siapa pun.
4. **Pengaturan → Push notification**: unggah kunci Firebase (project Firebase aplikasi OLT), bila push dipakai.
5. **Pengaturan → Akun CS**: buat akun staf lain.
6. **Simpan di luar server** (password manager / brankas): `docker/.env`, folder `docker/secrets/`, password backup, salinan `pgsodium_root.key` (backup).
7. **Data untuk tim aplikasi OLT**: `supabaseUrl` = `https://<domain API>`, `anonKey` = hasil `make anon-key`, versi library `v1.8.0` (panduan integrasi library).
8. Uji dari aplikasi: login, buka feed, kirim chat antar dua akun.

6. Bila ada masalah saat instalasi

Gejala	Penyebab umum	Tindakan
"does not resolve to any IP yet"	A record belum dibuat atau belum menyebar	Minta tim IT cek DNS, tunggu beberapa menit, jalankan ulang
Health tidak menjawab, `make logs s=traefik` berisi error ACME	Port 80 tertutup, DNS salah, atau batas Let's Encrypt	Pastikan 80/443 terbuka dari internet dan DNS benar, lalu `make up`
`docker/.env still contains CHANGE_ME` atau `is required`	Isian `.env` kurang	Jalankan ulang script, atau edit `docker/.env` lalu `make up`
`JWKS rejected`	Kunci dev, kunci privat, atau format salah	Minta ulang JWKS **publik** dari tim OLT
Clone terus gagal	Deploy key belum ditambahkan / repo lain	Periksa GitHub → Settings → Deploy keys. Kunci `github_deploy` untuk `chat_service_supabase`, kunci `github_deploy_hub` untuk `rendezvous_hub`
`Rendezvous Hub checkout not found at ...`	Domain Hub diisi tetapi `~/rendezvous_hub` belum ada	Jalankan ulang script (langkah 7), atau isi `HUB_DIR` di `docker/.env`
Hub 403 dari kantor	IP kantor tidak ada di `VPN_SOURCE_RANGES`	Cek IP publik kantor, perbaiki `docker/.env`, `make up`
`olt_mode` bukan `jwt`	Konfigurasi auth-bridge	`make logs s=auth-bridge`
Login dari aplikasi selalu "Sesi berakhir"	`iss` / `aud` /jam server OLT tidak cocok	`docker logs olt-social-auth-bridge-1 --since 15m` (field `reason`)

7. Belum ada JWKS OLT: mode dev sementara

Mode ini memakai OLT dummy (login uji `budi_santoso` / `olt12345` dan lainnya), tanpa domain dan HTTPS, diakses lewat SSH tunnel. Langkah 1–5 script tetap bisa dipakai: jalankan script, lalu berhenti (Ctrl+C) saat ditanya domain. Sesudahnya:

```
cd ~/chat_service_supabase
make up DUMMY_OLT=1
make seed
make smoke-dummy        # semua ok
make cs-staff           # akun staf dev untuk Hub
```

Hub di mode dev butuh clone `rendezvous_hub` di `~/rendezvous_hub` (tanpa itu `make up` berjalan tanpa `cs-console`). Karena langkah 7 script hanya berjalan bila domain Hub diisi, clone manual dengan deploy key kedua:

```
ssh-keygen -q -t ed25519 -N "" -C "deploy-hub@$(hostname)" -f ~/.ssh/github_deploy_hub
printf 'Host github-hub\n  HostName github.com\n  IdentityFile ~/.ssh/github_deploy_hub\n
IdentitiesOnly yes\n' >> ~/.ssh/config
cat ~/.ssh/github_deploy_hub.pub     # tambahkan di repo rendezvous_hub -> Settings -> Deploy keys
(baca saja)
git clone git@github-hub:dayat123123/rendezvous_hub.git ~/rendezvous_hub
cd ~/chat_service_supabase && make cs-console-react
```

Dari laptop:

```
ssh -L 8000:127.0.0.1:8000 -L 8090:127.0.0.1:8090 -L 8085:127.0.0.1:8085 <user>@<IP-server>
```

Selama terminal itu terbuka: Hub `http://localhost:8085`, API `http://localhost:8000`, login OLT dummy `http://localhost:8090`. Setelah JWKS OLT diterima: `make down`, lalu jalankan script lagi dan lanjutkan sampai selesai.

8. Sehari-hari

```
cd ~/chat_service_supabase
make ps                 # status
make logs s=rest        # log satu layanan (Ctrl+C keluar)
git pull && make up     # update versi, di luar jam bursa, setelah make backup
```

Bila Hub dipasang, perbarui juga repo Hub sebelum `make up`: `git -C ~/rendezvous_hub pull --ff-only` (atau jalankan ulang script, yang menarik kedua repo).

`PROFILE=single` (dan `CS=1` bila Hub dipasang) sudah disimpan script di `~/.bashrc`, jadi perintah `make` langsung memakai profil produksi. Jangan pernah menjalankan `make reset` atau `docker volume rm` di server ini.

Lampiran — Isi script `install.sh`

> **Untuk dibaca, bukan disalin.** Baris panjang di bawah terlipat agar muat di kertas; bila disalin dari PDF, lipatannya bisa merusak script. Kirim file aslinya: `scp scripts/server-install.sh <user>@<IP-server>:~/install.sh`

```
#!/usr/bin/env bash
set -euo pipefail

REPO_SSH="${REPO_SSH:-git@github.com:dayat123123/chat_service_supabase.git}"
APP_DIR="${APP_DIR:-$HOME/chat_service_supabase}"
HUB_REPO_SSH="${HUB_REPO_SSH:-git@github-hub:dayat123123/rendezvous_hub.git}"
HUB_APP_DIR="${HUB_APP_DIR:-$HOME/rendezvous_hub}"
SCRIPT_PATH="$(readlink -f "$0")"

say() { printf '\n\033[1;34m==> %s\033[0m\n' "$*"; }
ok() { printf '\033[32m  ok  %s\033[0m\n' "$*"; }
warn() { printf '\033[33m  !   %s\033[0m\n' "$*"; }
die() { printf '\033[31m  x   %s\033[0m\n' "$*" >&2; exit 1; }
ask() {
  local prompt=$1 default=${2:-} reply
  read -r -p "  $prompt${default:+ [$default]}: " reply
  printf '%s' "${reply:-$default}"
}
confirm() {
  local reply
  read -r -p "  $1 [y/N]: " reply
  [[ $reply =~ ^[Yy]$ ]]
}
env_get() {
  [ -f "$APP_DIR/docker/.env" ] || return 0
  sed -n "s/^$1=//p" "$APP_DIR/docker/.env" | tail -1
}
env_set() {
  local file="$APP_DIR/docker/.env" tmp
  tmp=$(mktemp)
  awk -v k="$1" -v v="$2" '
    $0 ~ "^"k"=" { if (!done) print k"="v; done = 1; next }
    { print }
    END { if (!done) print k"="v }
  ' "$file" > "$tmp"
  cat "$tmp" > "$file"
  rm -f "$tmp"
}
bashrc_export() {
  grep -qx "export $1=$2" "$HOME/.bashrc" 2>/dev/null || echo "export $1=$2" >> "$HOME/.bashrc"
}

[ "$(id -u)" -ne 0 ] || die "Run as a regular user with sudo, not as root."
. /etc/os-release
[ "${ID:-}" = ubuntu ] || die "This script is for Ubuntu (detected: ${ID:-unknown})."
sudo -v

say "1/10 System packages, automatic security updates, clock"
sudo apt-get update -qq
```

```
sudo DEBIAN_FRONTEND=noninteractive apt-get -y -qq upgrade
sudo DEBIAN_FRONTEND=noninteractive apt-get -y -qq install git make curl ca-certificates unattended-upgrades
ufw python3
echo 'unattended-upgrades unattended-upgrades/enable_auto_updates boolean true' | sudo debconf-set-selections
sudo DEBIAN_FRONTEND=noninteractive dpkg-reconfigure -f noninteractive unattended-upgrades
sudo timedatectl set-ntp true
ok "packages installed, automatic security updates on, NTP on"

say "2/10 Docker"
if ! command -v docker >/dev/null 2>&1; then
  sudo install -m 0755 -d /etc/apt/keyrings
  sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
  echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu ${VERSION_CODENAME} stable" \
    | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
  sudo apt-get update -qq
  sudo DEBIAN_FRONTEND=noninteractive apt-get -y -qq install docker-ce docker-ce-cli containerd.io docker-
buildx-plugin docker-compose-plugin
fi
sudo systemctl enable --now docker > /dev/null
if ! id -nG "$USER" | grep -qw docker; then
  sudo usermod -aG docker "$USER"
fi
if ! docker info > /dev/null 2>&1; then
  ok "user $USER added to the docker group; continuing with the new group"
  exec sg docker -c "bash '$SCRIPT_PATH'"
fi
compose_version=$(docker compose version --short | sed 's/^v//')
printf '%s\n2.24.0\n' "$compose_version" | sort -V -C -r || die "docker compose $compose_version is too old
(needs 2.24 or newer)."
ok "docker $(docker version --format '{{.Server.Version}}'), compose $compose_version"

say "3/10 Swap (only when RAM < 8 GB)"
mem_kb=$(awk '/MemTotal/ {print $2}' /proc/meminfo)
if [ "$mem_kb" -lt 7500000 ] && ! swapon --show | grep -q .; then
  sudo fallocate -l 4G /swapfile
  sudo chmod 600 /swapfile
  sudo mkswap /swapfile > /dev/null
  sudo swapon /swapfile
  grep -q '^/swapfile ' /etc/fstab || echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab > /dev/null
  ok "4 GB swap created"
else
  ok "no extra swap needed"
fi

say "4/10 Server firewall (ufw, extra layer)"
ssh_port=$(sudo sshd -T 2>/dev/null | awk '/^port / && !seen {print $2; seen = 1}' || true)
ssh_port=${ssh_port:-22}
sudo ufw allow "$ssh_port/tcp" > /dev/null
sudo ufw allow 80/tcp > /dev/null
sudo ufw allow 443/tcp > /dev/null
sudo ufw --force enable > /dev/null
ok "ufw: $ssh_port (SSH), 80, 443"
warn "Docker container ports bypass ufw; the main restriction stays in the hosting provider's firewall"

say "5/10 Code from GitHub"
if [ -d "$APP_DIR/.git" ]; then
  git -C "$APP_DIR" pull --ff-only
  ok "code updated in $APP_DIR"
```

```
else
  mkdir -p "$HOME/.ssh"
  chmod 700 "$HOME/.ssh"
  [ -f "$HOME/.ssh/github_deploy" ] || ssh-keygen -q -t ed25519 -N "" -C "deploy@$(hostname)" -f
"$HOME/.ssh/github_deploy"
  grep -q 'IdentityFile ~/.ssh/github_deploy' "$HOME/.ssh/config" 2>/dev/null \
    || printf 'Host github.com\n  IdentityFile ~/.ssh/github_deploy\n  IdentitiesOnly yes\n' >>
"$HOME/.ssh/config"
  chmod 600 "$HOME/.ssh/config"
  ssh-keygen -F github.com > /dev/null 2>&1 || ssh-keyscan -t ed25519 github.com >> "$HOME/.ssh/known_hosts"
2>/dev/null
  until git clone -q "$REPO_SSH" "$APP_DIR" 2>/dev/null; do
    echo
    echo "  Add this deploy key on GitHub:"
    echo "  repo chat_service_supabase -> Settings -> Deploy keys -> Add deploy key"
    echo "  (any title, do NOT tick \"Allow write access\")"
    echo
    cat "$HOME/.ssh/github_deploy.pub"
    echo
    read -r -p "  Press Enter once the deploy key is added... " _
  done
  ok "code in $APP_DIR"
fi
cd "$APP_DIR"

say "6/10 Configuration (docker/.env)"
make -s gen-secrets
domain=$(ask "API domain (e.g. social-api.company.co.id)" "$(env_get PROXY_DOMAIN)")
[[ $domain =~ ^[A-Za-z0-9.-]+\.[A-Za-z]{2,}$ ]] || die "Invalid domain: $domain"
acme_email=$(ask "IT team email for HTTPS certificates" "$(env_get ACME_EMAIL)")
[[ $acme_email == *@*.* ]] || die "Invalid email: $acme_email"
hub_domain=$(ask "Rendezvous Hub domain (leave empty if none yet)" "$(env_get CS_CONSOLE_DOMAIN)")
vpn_ranges=""
if [ -n "$hub_domain" ]; then
  [[ $hub_domain =~ ^[A-Za-z0-9.-]+\.[A-Za-z]{2,}$ ]] || die "Invalid Hub domain: $hub_domain"
  vpn_ranges=$(ask "IPs/ranges allowed to open the Hub, comma separated (e.g. 203.0.113.10/32)" "$(env_get
VPN_SOURCE_RANGES)")
  [[ $vpn_ranges =~ ^[0-9./,\ ]+$ ]] || die "Invalid Hub IP list: $vpn_ranges"
  [[ $vpn_ranges != *0.0.0.0/0* ]] || die "0.0.0.0/0 opens the Hub to the whole internet. Enter office/VPN IPs
only."
fi
backup_default=$(env_get BACKUP_LOCAL_REPO_DIR)
[ -n "$backup_default" ] || { [ -d /mnt/backup ] && backup_default=/mnt/backup/restic; }
backup_dir=$(ask "Local backup folder on a separate disk (empty = backups/restic in the repo)"
"$backup_default")

env_set PROXY_DOMAIN "$domain"
env_set SUPABASE_PUBLIC_URL "https://$domain"
env_set API_EXTERNAL_URL "https://$domain/auth/v1"
env_set ACME_EMAIL "$acme_email"
env_set HEALER_DOCKER_GID "$(getent group docker | cut -d: -f3)"
if [ -n "$hub_domain" ]; then
  env_set CS_CONSOLE_DOMAIN "$hub_domain"
  env_set VPN_SOURCE_RANGES "$vpn_ranges"
fi
if [ -n "$backup_dir" ]; then
  [[ $backup_dir == /* ]] || die "The backup folder must be an absolute path: $backup_dir"
  sudo install -d -o 10001 -g 10001 -m 700 "$backup_dir"
  env_set BACKUP_LOCAL_REPO_DIR "$backup_dir"
```

```
fi
ok "docker/.env updated"

say "7/10 Rendezvous Hub code (only when the Hub is installed)"
if [ -d "$HUB_APP_DIR/.git" ]; then
  git -C "$HUB_APP_DIR" pull --ff-only
  ok "Hub code updated in $HUB_APP_DIR"
elif [ -n "$hub_domain" ]; then
  mkdir -p "$HOME/.ssh"
  chmod 700 "$HOME/.ssh"
  [ -f "$HOME/.ssh/github_deploy_hub" ] || ssh-keygen -q -t ed25519 -N "" -C "deploy-hub@$(hostname)" -f
"$HOME/.ssh/github_deploy_hub"
  grep -qx 'Host github-hub' "$HOME/.ssh/config" 2>/dev/null \
    || printf 'Host github-hub\n  HostName github.com\n  IdentityFile ~/.ssh/github_deploy_hub\n
IdentitiesOnly yes\n' >> "$HOME/.ssh/config"
  chmod 600 "$HOME/.ssh/config"
  ssh-keygen -F github.com > /dev/null 2>&1 || ssh-keyscan -t ed25519 github.com >> "$HOME/.ssh/known_hosts"
2>/dev/null
  until git clone -q "$HUB_REPO_SSH" "$HUB_APP_DIR" 2>/dev/null; do
    echo
    echo "  Add this second deploy key on GitHub (GitHub needs a separate deploy key per repo):"
    echo "  repo rendezvous_hub -> Settings -> Deploy keys -> Add deploy key"
    echo "  (any title, do NOT tick \"Allow write access\")"
    echo
    cat "$HOME/.ssh/github_deploy_hub.pub"
    echo
    read -r -p "  Press Enter once the deploy key is added... " _
  done
  ok "Hub code in $HUB_APP_DIR"
else
  ok "skipped (no Hub domain)"
fi
if [ -d "$HUB_APP_DIR/.git" ] && { [ -n "$(env_get HUB_DIR)" ] || [ "$HUB_APP_DIR" != "$(dirname
"$APP_DIR")/rendezvous_hub" ]; }; then
  env_set HUB_DIR "$HUB_APP_DIR"
  ok "docker/.env: HUB_DIR=$HUB_APP_DIR"
fi

say "8/10 OLT public keys (docker/secrets/olt_jwks)"
jwks_file="$APP_DIR/docker/secrets/olt_jwks"
jwks_check() {
  python3 - "$1" <<'PY'
import json, sys
try:
    keys = json.load(open(sys.argv[1]))["keys"]
except Exception as e:
    sys.exit(f"not a valid JWKS: {e}")
if not keys:
    sys.exit("the JWKS contains no keys")
for k in keys:
    if str(k.get("kid", "")).startswith("dev-"):
        sys.exit("still a dev key (kid starts with dev-); the production profile rejects it")
    if k.get("alg") not in ("EdDSA", "ES256") or "d" in k:
        sys.exit("keys must be public with alg EdDSA or ES256")
print(", ".join(k.get("kid", "?") for k in keys))
PY
}
if kids=$(jwks_check "$jwks_file" 2>/dev/null); then
  ok "OLT JWKS installed (kid: $kids)"
```

```
else
  echo "  Paste the JWKS (PUBLIC keys) from the OLT backend team, then press Enter and Ctrl+D:"
  tmp_jwks=$(mktemp)
  cat > "$tmp_jwks"
  if kids=$(jwks_check "$tmp_jwks"); then
    (umask 077; cat "$tmp_jwks" > "$jwks_file")
    rm -f "$tmp_jwks"
    ok "OLT JWKS saved (kid: $kids)"
  else
    rm -f "$tmp_jwks"
    die "JWKS rejected. Ask the OLT team for the public-key JWKS (docs/olt-integration.md), then run this
script again."
  fi
fi

say "9/10 DNS check"
check_dns() {
  local name=$1 resolved
  resolved=$(getent ahostsv4 "$name" | awk '!seen {print $1; seen = 1}' || true)
  [ -n "$resolved" ] || die "$name does not resolve to any IP yet. Ask the IT team to create an A record, wait,
then run again."
  echo "  $name -> $resolved   (IPs on this server: $(hostname -I | xargs))"
}
check_dns "$domain"
[ -z "$hub_domain" ] || check_dns "$hub_domain"
confirm "Is the IP above this server's public IP (or the office public IP forwarded to this server)?" \
  || die "Fix DNS first, then run this script again."

say "10/10 Starting services"
export PROFILE=single
bashrc_export PROFILE single
if [ -n "$hub_domain" ]; then
  export CS=1
  bashrc_export CS 1
fi
make up
ok "all services running"

if [ -n "$hub_domain" ] && confirm "Create the first Hub Super Admin account now?"; then
  admin_email=$(ask "Super Admin email")
  admin_name=$(ask "Super Admin full name")
  make cs-bootstrap-admin EMAIL="$admin_email" NAME="$admin_name" \
    || warn "account not created (a Super Admin may already exist). See the message above."
fi

say "Public access check"
anon_key=$(cat docker/secrets/anon_key)
health=""
for _ in $(seq 1 18); do
  health=$(curl -fsS -m 10 -H "apikey: $anon_key" "https://$domain/bridge/v1/health" 2>/dev/null || true)
  [[ $health == *'"olt_mode":"jwt"'* ]] && break
  sleep 10
done
[[ $health == *'"olt_mode":"jwt"'* ]] && ok "https://$domain/bridge/v1/health: olt_mode jwt" \
  || warn "health is not answering yet. Check the certificate: make logs s=traefik (DNS/port 80 must be open)"
code=$(curl -s -o /dev/null -m 10 -w '%{http_code}' "http://$domain/" || true)
[[ $code =~ ^30[1278]$ ]] && ok "http redirects to https ($code)" || warn "http://$domain/ answered $code
(expected a redirect)"
code=$(curl -s -o /dev/null -m 10 -w '%{http_code}' "https://$domain/" || true)
```

```
[ "$code" = 404 ] && ok "https://$domain/ 404 (only API paths are open)" || warn "https://$domain/ answered
$code (expected 404)"

cat <<EOF

Done.

  URL API (supabaseUrl) : https://$domain
  Anon key (anonKey)    : $anon_key
EOF
[ -z "$hub_domain" ] || echo "  Rendezvous Hub        : https://$hub_domain   (only from: $vpn_ranges)"
cat <<EOF

Next steps (docs/server-install.md section 5):
  1. Log in to the Hub as Super Admin and change the temporary password.
  2. Pengaturan -> Backup: choose a destination, create an encryption password, store it off the server.
  3. Pengaturan -> Alert: fill in Telegram/email, then Kirim uji.
  4. Store off the server: docker/.env, docker/secrets/, the backup password.
  5. Give the API URL and anon key to the OLT app team.
EOF
```